

Results After β -Irradiation ($^{106}\text{Ru}/^{106}\text{Rh}$) of Choroidal Melanomas

Twenty Years' Experience

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For two decades, from 1964 to 1984, 309 patients suffering from choroidal melanoma were treated with $^{106}\text{Ru}/^{106}\text{Rh}$ applications, following confirmation of diagnosis by a variety of tests. A total of 216 patients (69.9%) were treated successfully and have been under observation for a mean period of 6.7 years after irradiation. 188 patients were followed for more than five years. Results indicate that enucleation for choroidal melanoma, especially in cases of eyes with good vision, may no longer be the standard treatment for this disease. This appears particularly true inasmuch as almost 50% of all patients with large choroidal melanomas who have enucleation die from metastases within 5 years of the operation. Therefore, conservative methods such as photocoagulation, irradiation, and microsurgical excision have been used with more or less success to destroy tumor and save a functioning eye.

Treatment for choroidal melanoma by enucleation of the eye, especially one with good vision, may no longer be standard, principally because almost 50% of all patients with this disease who have enucleations will die from metastases within 5 years after the operation. Therefore, conservative methods such as photocoagulation, irradiation, and microsurgical excision have been used with more or less success to destroy the tumor and to save a functioning eye. By treating patients successfully with ^{60}Co applicators, Stallard (1) proved that the earlier assumption of absolute radioresistance of choroidal melanomas could not be maintained.

To reduce the severe radiogenic side-effects, we tested other radionuclides and $^{106}\text{Ru}/^{106}\text{Rh}$ was found to be a suitable energy source to treat intraocular tumors (2-5).

PATIENTS AND METHODS

Between 1964 and 1984, a total of 309 patients (168 men, 141 women) suffering from choroidal melanoma were treated with $^{106}\text{Ru}/^{106}\text{Rh}$ applicators after diagnosis has been confirmed by ophthalmoscopy, ultrasonography, fluorescein angiography, and ^{32}P test. Biopsy of intraocular tissue seems hazardous and was not carried out. The age distribution is shown in Table 1.

We use concave mirror-like applicators consisting of pure sheet silver 1.0 mm thick which contain $^{106}\text{Ru}/^{106}\text{Rh}$ in equal distribution. The front window on the concave side is 0.1 mm and allows the electrons to pass practically unhindered. The back is 0.9 mm thick and absorbs nearly 95% of the beta irradiation. These ophthalmic applicators have two ears with the help of

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TABLE 1. Age distribution of patients suffering from choroidal melanoma

Age (yr)	No. of patients
10-20	4
20-30	11
30-40	22
40-50	88
50-60	81
60-70	72
70-80	29
80-90	2
Total	309

which they can be sutured to the sclera. Several types of Ruthenium applicators are available (Fig. 1). Physical properties, experimental studies, and surgical procedures were published earlier (3).

The tumor-damaging dose—following Stallard's recommendation—was at least 100 Gy (10,000 rad) at the apex of the tumor. Thus, values above 1,000 Gy (100,000 rad) may be reached at the tumor base. An irradiation time of 8–14 days seemed to be the optimum exposure time.

After treatment, the patients were examined every month and later, if the tumor began to shrink, only every 3 months. Tumor regression was documented by fundus photography, ultrasonography, and fluorescein angiography.

The average follow-up period of all cases was 6.7 years (shortest period, 1 year; longest, 21 years); 188 patients could be followed up for more than 5 years (average: 9.4 years); and 87 patients were controlled for more than 10 years.

RESULTS

In this study, those patients whose tumors had either changed to a flat scar or had shrunk to a still appreciable grayish or black mass with scarring of the tissue around the tumor, which had remained unchanged for more than 1 year, were considered to have been treated successfully. Moreover, those patients who died later from causes other than metastases and whose choroidal melanoma had shrunk or cicatrized were also included as successfully treated cases. Only those patients who died from metastases or who had to be enucleated were counted as failures.

Two hundred sixteen (69.9%) of 309 (100%) patients could be treated successfully and have been under observation since then, for a mean period of 6.7 years after irradiation (shortest period, 1 year; longest, 21 years). Fifty-three (17.2%) patients had to be enucleated because of tumor new growth, but did not die from metastases, and 40 (12.9%) died from proved metastases in this period of 20 years (Fig. 2).

One hundred eighty-eight patients could be followed up for more than 5 years (101 patients for 5–9 years, 63 patients for 10–14 years, and 24 patients for 15–21 years). Out of the group with long-term follow-up for a mean period of 9.4 years, 130 (69.1%) patients could be regarded as successfully treated cases, 40 (21.3%) had to be enucleated and are either alive or died from other causes than metastases, and 18 patients died from proven metastases (Fig. 3).

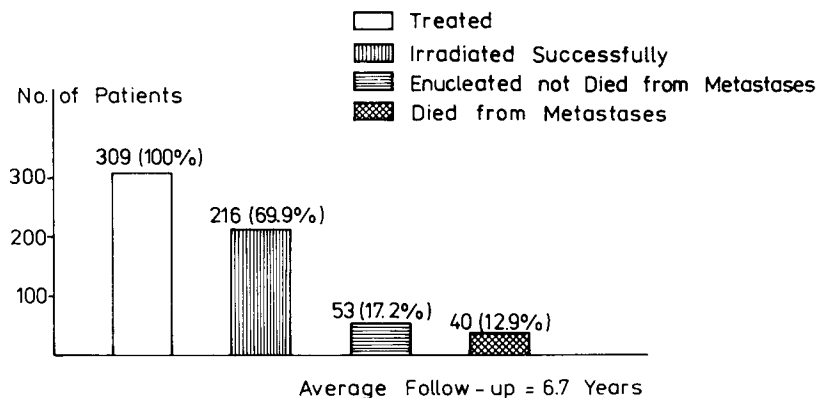
Tumor Regression

One hundred fourteen (52.8%) of the successfully treated patients had a flat scar after therapy (Fig. 4).

FIG. 1. $^{106}\text{Ru}/^{106}\text{Rh}$ eye applicators (Isocommerz GmbH, Berlin-Buch, German Democratic Republic).



FIG. 2. Results after β -irradiation with $^{106}\text{Ru}/^{106}\text{Rh}$ plaques of 309 patients.



The other eyes showed more or less notable remnants of the former tumor. The fact that tumor regression after this kind of therapy may take many months is important for avoiding any precipitous enucleation. The regression to a flat pigment stippled scar depends primarily on tumor size. However, all tumors larger than 3 mm in height left behind an elevated mass, which should be observed carefully during the follow-up period because of some possible reactivation process (Fig. 5).

Visual Acuity

Preservation of useful visual acuity is a principal aim of beta-ray treatment. Forty-nine (22.7%) of all successfully treated patients retained good visual acuity between 1.5 and 0.5. About half the patients (54.2%) had only a poor central visual acuity of less than 0.2 because the macula had been involved in the cicatricial formation after treatment. Visual field deficiencies corresponding to the retinochoroidal scar could be registered in all our patients.

Tumor Size

Two hundred nineteen (70.9%) of the patients had small tumors, each with its greatest dimension not more than 10 mm with an elevation not more than 3 mm (T 1a, T 1b). Twenty-six (8.4%) patients suffered from

medium-sized tumors not more than 15 mm in its greatest dimension and with an elevation not more than 5 mm (T 2). Sixty-four (20.7%) had large melanomas more than 15 mm in its greatest dimension or with an elevation of more than 5mm (T 3). The prevalence of small tumors (81.5%) is even clearer in the group of the 216 successfully treated cases (Table 2). The group of 40 patients dying from metastases showed a prevalence of large tumors (42.5%) (Table 3).

β -Irradiation and Xenon Coagulation

Light coagulation was additionally performed after β -irradiation in 34 cases to occlude the choroidal vessels around the tumor. First, the central rim of the tumor was surrounded by coagulations. Later, if the tumor was not completely destroyed, the surface of the tumor proper was also coagulated. Especially in eyes with the melanoma close to the optic disc and where the applicator cannot be placed completely around the tumor, we succeeded in partially surrounding the central tumor rim with coagulation effects. The results obtained in our cases with additional light coagulation are shown in Table 4.

Radiogenic Side-Effects

Radiogenic tissue damage must be expected after each kind of radiotherapy, particularly the serious late

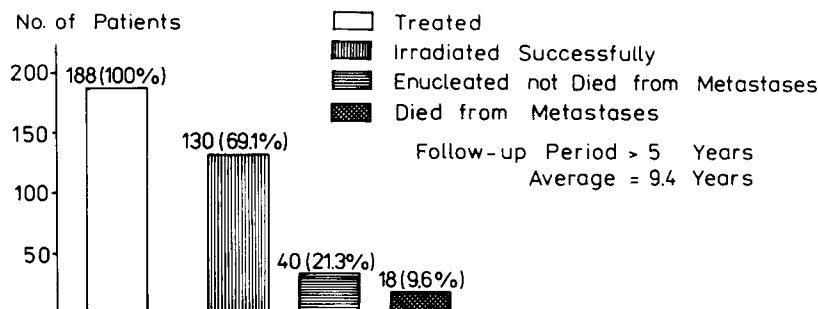
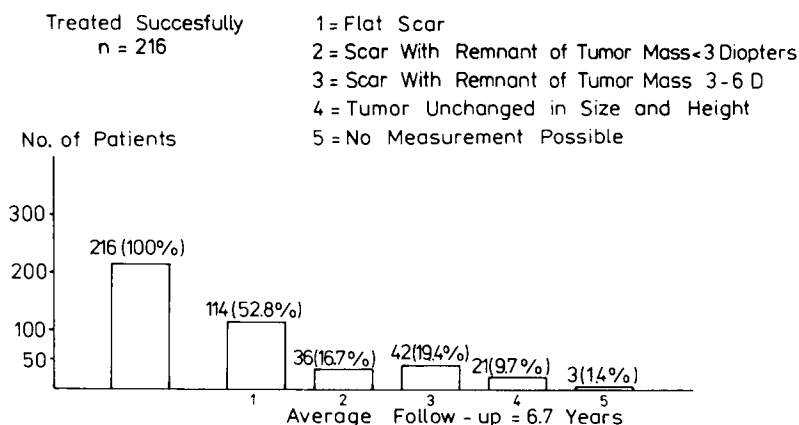


FIG. 3. Long-term results after β -irradiation with $^{106}\text{Ru}/^{106}\text{Rh}$ plaques of 188 patients.

FIG. 4. Tumor regression after β -irradiation.



effects which may occasionally occur after β -irradiation. Exudative reactions occurring a few days after irradiation, such as chemosis, choroidal detachment, and even transient retinal detachment, should not be regarded as serious complications because of their good prognosis. Against earlier expectations, the primary concern is now the postradiation retinopathy that has frequently occurred after β -irradiation, sometimes even years after initially successful treatment.

Radiogenic complications which led to deterioration of visual acuity are shown in Table 5. Most of the patients—83 patients—with a tumor located 1 to 2 disc diameters away from the posterior pole showed a destruction of the macula because the scarring around the tumor necessarily involved this retinal area.

Enucleation After β -Irradiation

Enucleation was necessary in 64 cases, mostly within the first year after treatment. The reason for removing the eye was new tumor growth in 54 and loss of visual

acuity due to radiogenic side-effects in 10 cases. Histologic examination of the enucleated eyes showed active new-grown melanoma tissue containing some tumor cells in four, scar tissue without tumor cells in four, and no histologic examination was available in two cases.

Of all 64 enucleated patients, 11 (17.2%) died from metastases within 1-7 years after enucleation, and nine patients died from other causes.

Deaths After β -Irradiation

Seventy-five patients of this series died during the follow-up period: 40 of metastases, 27 of diseases other than melanoma (cardiac infarction or apoplectic stroke, 19; other malignant tumors, six; cirrhosis of liver, one; suicide, one), and eight of unknown causes. The death from metastases had been verified by autopsy in 19, laparoscopy in four, and by clinical examination in 17 cases. Metastases occurred mainly in the liver, where in some cases an extreme hepatomegaly had been produced.

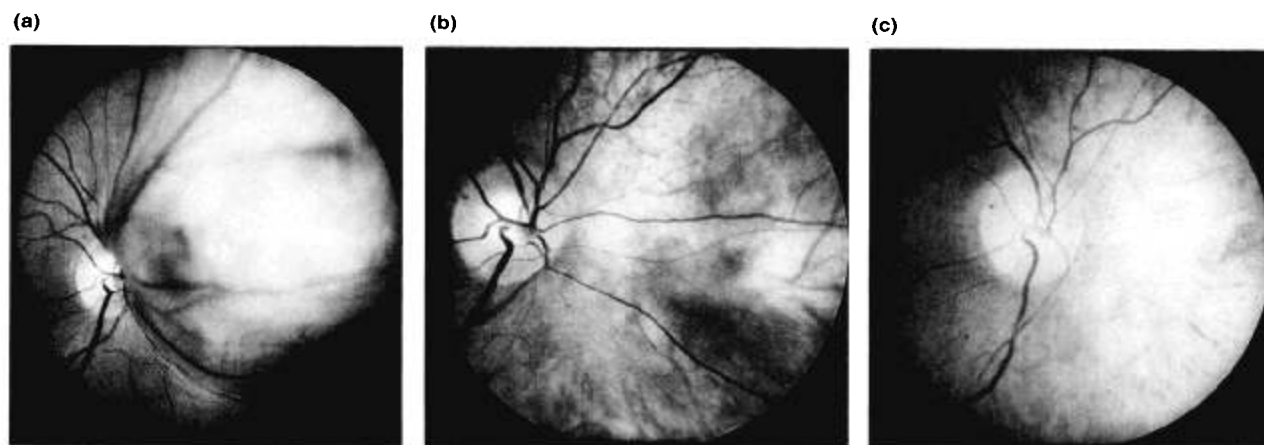


FIG. 5. **A:** Choroidal melanoma of right eye before treatment. **B:** Same eye 2 years after β -irradiation with $^{106}\text{Ru}/^{106}\text{Rh}$ 1,084 Gy within 14 days at scleral surface. **C:** Same eye 8 years after irradiation. Visual acuity 0.5.

TABLE 2. Prevalence of small tumors in 216 successfully treated cases

No. of patients	(%)	Size of tumor
176	(81.5)	Small
13	(6.0)	Medium
27	(12.5)	Large

Survival Rate After $^{106}\text{Ru}/^{106}\text{Rh}$ Therapy

The survival rate was calculated using the life-table method. As shown in Fig. 6, survival rates of 84.3% (deaths from any causes) or 88.7% (deaths from metastases only) after 5 years, and 65.8% (deaths from any causes) or 79.5% (deaths from metastases only) after 10 years could be achieved. In this diagram, the survival rates of patients who died from metastases and those who died from any causes were plotted separately. A comparison with results published in previous studies (6) shows that survival after local β -irradiation is better than after enucleation of eyes containing choroidal melanomas similar in size and extension.

DISCUSSION

More recently, the value of enucleation has been questioned, and some authorities have even pointed out that this procedure may endanger our patient's life (7,8). This hypothesis has provoked objections (9–11), so that additional studies will be needed to determine whether our conception of the management of choroidal melanoma is correct.

Radiotherapy has become more accepted recently, especially the use of radioactive plaques and charged particle irradiation (12–14). The results presented in this study lead us to the conclusion that beta-irradiation using $^{106}\text{Ru}/^{106}\text{Rh}$ plaques is an effective procedure, especially in treating small and medium-sized choroidal melanomas. A successful therapeutic outcome in about two-thirds of the cases can be achieved, provided that the following recommendations are observed: first, be-

TABLE 3. Prevalence of large tumors in 40 patients dying from metastases

No. of patients	(%)	Size of tumor
18	(45.0)	Small
5	(12.5)	Medium
17	(42.5)	Large

TABLE 4. Results of 34 cases treated with additional light coagulation

Status of patients	No. of patients
Treated successfully	21
Living, but enucleated	4
Dying from metastases	9
Total	4
	34

cause of the physical properties of $^{106}\text{Ru}/^{106}\text{Rh}$, the choroidal melanoma should not exceed 7 mm in height above the scleral surface. Second, distance of the posterior margin should be at least 1 to 2 disc diameters from the optic disc to avoid radiogenic damage of the optical nerve. Third, there should be no extended tumor growth outside the eye.

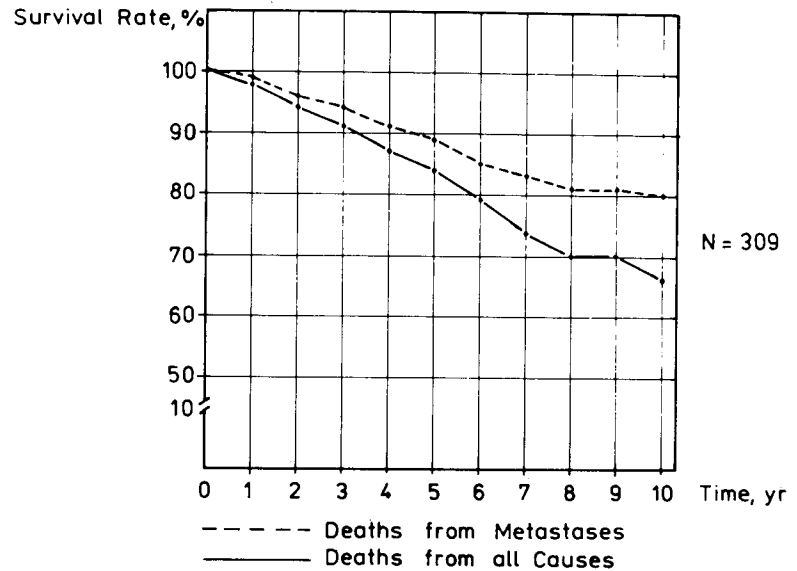
Local irradiation with $^{106}\text{Ru}/^{106}\text{Rh}$ plaques is not likely to result in a higher incidence of metastases than enucleation, although the only scientific way to answer this question would be to conduct a randomized prospective study. Rotman et al. (15) pointed out that those methods of treatment which left the eye in situ had a greater overall survival rate than enucleation. Maybe irradiation, in contrast to enucleation, promotes an immune response that may indeed control distal metastasis.

The management of choroidal melanomas will probably remain controversial. New radiotherapeutic procedures such as ^{125}I seeds (12), proton-beam irradiations (13), or heavy helium ions (14) have been published recently, but long-term follow-up examinations are still lacking. ☉

TABLE 5. Radiogenic complications in patients leading to deterioration of visual acuity

Complication	No. of patients
Macular destruction because of scarring around the tumor	83
Atrophy of the optic nerve	23
Macular degeneration	16
Postradiation retinopathy with edema of the optic disc	17
Partial cuneiform cataract without loss of vision	5
Total cataract	7
Vitreous hemorrhage	10
Secondary glaucoma	3
Thrombosis of the central retinal vein	2
Scleral necrosis	1
Exudative reactions such as choroidal and retinal detachment with spontaneous improvement	21

FIG. 6. Survival rate of patients (N = 309) with choroidal melanoma treated with $^{106}\text{Ru}/^{106}\text{Rh}$ β -ray applicator. Broken line indicates deaths from metastases; solid line, deaths from any causes.



REFERENCES

1. Stallard HB. Radiotherapy for malignant melanoma of the choroid. *Br J Ophthalmol* 1966;50:147-55.
2. Lommatzsch PK. Beta-ray treatment of malignant epibulbar melanoma. *Graefes Arch Klin Exp Ophthalmol* 1978;209:111-24.
3. Lommatzsch PK. Treatment of choroidal melanomas with $\text{Ru-}^{106}/\text{Rh-}^{106}$ applicators. *Surv Ophthalmol* 1974;19:85-100.
4. Lommatzsch PK. Radiotherapie der intraokularen Tumoren, insbesondere bei Aderhautmelanom. *Klin Monatsbl Augenheilkd* 1979;174:948-58.
5. Lommatzsch PK. β -irradiation of choroidal melanoma with $^{106}\text{Ru}/^{106}\text{Rh}$ applicators. *Arch Ophthalmol* 1983;101:713-7.
6. Kiehl H, Kirsch I, Lommatzsch P. Das Überleben nach Behandlung des malignen Melanoms der Aderhaut: Vergleich von konservativer Therapie ($^{106}\text{Ru}/^{106}\text{Rh}$ Applikator) und Enukleation ohne und mit postoperativer Orbitabestrahlung, 1960 bis 1979. *Klin Monatsbl Augenheilkd* 1984;184:2-14.
7. Zimmerman LE, McLean IW, Foster WD. Does enucleation of the eye containing a malignant melanoma prevent or accelerate the dissemination of tumor cells? *Br J Ophthalmol* 1978;62:420-5.
8. Zimmerman LE, McLean IW. Effect of enucleation on uveal melanomas. *Am J Ophthalmol* 1979;87:741-60.
9. Boniuk M. A crisis in the management of patients with choroidal melanoma. *Am J Ophthalmol* 1979;87:840-2.
10. Jakobiec FA. A moratorium on enucleation for choroidal melanoma. *Am J Ophthalmol* 1979;87:842-6.
11. Manschot WA, van Peperzeel HA. Choroidal melanoma. Enucleation or observation? A new approach. *Arch Ophthalmol* 1980;98:71-7.
12. Packer S, Rotman M, Fairchild RG, et al. Irradiation of choroidal melanoma with iodine 125 ophthalmic plaque. *Arch Ophthalmol* 1980;98:1453-7.
13. Gragoudas ES, Goitein M, Verhey L, et al. Proton beam irradiation: an alternative to enucleation for intraocular melanomas. *Ophthalmology* 1980;87:571-81.
14. Char DH, Castro JR, Quivey JM, et al. Helium ion charged particle therapy for choroidal melanoma. *Ophthalmology* 1980;87:565-70.
15. Rotman M, Long RS, Packer S, et al. Radiation therapy of choroidal melanoma. *Trans Ophthalmol Soc UK* 1977;97:431-5.